Zinc Metal-Promoted Nucleophilic Addition of Azetidin-2-ones to Aldehydes and Nitriles

Fides Benfatti, Giuliana Cardillo,* Serena Fabbroni, Luca Gentilucci, Rossana Perciaccante, Fabio Piccinelli, Alessandra Tolomelli

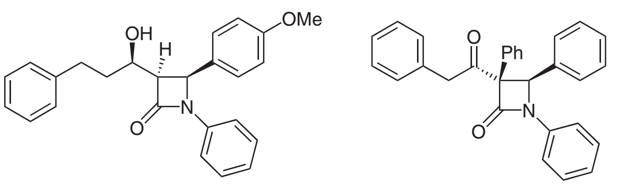
Dipartimento di Chimica 'G. Ciamician', Università di Bologna, Via Selmi 2, 40126 Bologna, Italy
Fax +39(051)2099456; E-mail: giuliana.cardillo@unibo.it

Received 27 July 2004; revised 6 September 2004

Abstract: The reactivity of in situ generated organozinc reagents of 3-alkenyl-3-bromoazetidin-2-ones with aromatic and aliphatic aldehydes to give the corresponding alcohol derivatives has been studied. The effect of solvent and Lewis acids has been explored in order to improve the reaction yields. The application of this methodology to nitriles and acyl chlorides allowed the preparation of the corresponding keto derivatives. The products of these reactions could be of interest on account of their structural similarity with the already known cholesterol adsorption inhibitors.

Key words: zinc, aldehydes, condensation, azetidinones, allylic organozinc reagents, Lewis acids

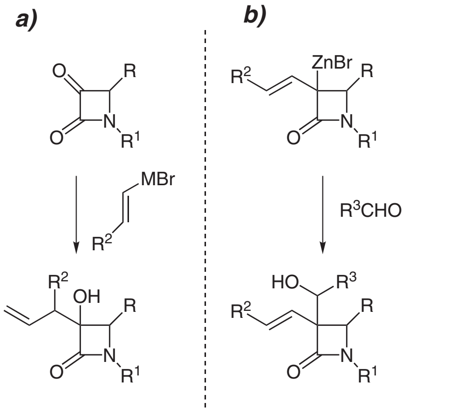
Organozinc reagents are very useful and versatile compounds for the carbon–carbon bond formation.[1] The Reformatsky and Barbier reactions[2] have been widely used due to the easy availability of these carbanions through the oxidative addition of metallic zinc to activated organic halides such as α-halo esters, α-halo amides and allyl halides. Recently, a class of β-lactam based structures, containing hydroxy and keto functions in the C-3 side chain, were found to be very potent cholesterol adsorption inhibitors (CAI) (Figure 1).[3]



SCH 56524

Figure 1 Representative examples of cholesterol adsorption inhibitors containing hydroxy and keto functions in the side carbon chain.

In a program directed towards the synthesis of potential protease inhibitors, we became interested in the introduction of oxygenated substituents at the C-3 position of 3-alkenylazetidin-2-ones. In most cases, azetidin-2,3-diones have been regarded as starting materials for the introduction of functionalized side chain at C-3 position via condensation with organometallic allylic reagents (Scheme 1, **a**).[4] Following a different approach, we investigated the regio- and stereochemistry of the nucleophilic attack of C-3 zinc organometallic derivatives, obtained from α-bromo- α-alkenyl-β-lactams, on aldehydes and nitriles (Scheme 1, **b**).



Scheme 1 Different reaction pathways for the synthesis of hydroxyazetidinone derivatives.

The starting 3-alkenyl-3-bromoazetidin-2-ones **1** were directly prepared in a one-pot synthesis through the Staudinger reaction[5] of 2-bromoalkenyl chlorides[6] and the appropriate imines with triethylamine as reported by Bose and Manhas for the synthesis of α-vinyl-β-lactams.[7] The β-lactams could be obtained in 50–60% yields, preferentially in the *cis* configuration. Recently, densely substituted β-lactams have been recognized as useful intermediates for a variety of further transformations as well as biologically active molecules. To the best of our knowledge, among the examples of substituted β-lactams,[3,8] 3-alkenyl-3-bromoazetidin-2-ones have not been described.

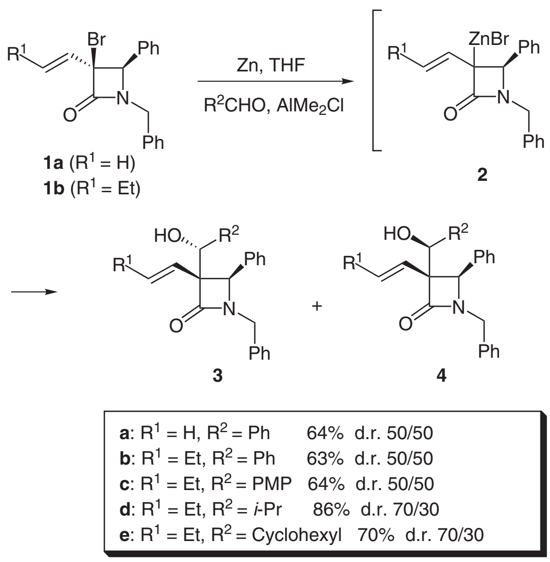
In the course of our investigation on the reactivity of this class of compounds, the easy preparation of organozinc derivatives **2** and their reactivity with aldehydes and nitriles was studied (Scheme 2). Although the transformation of C-3 substituted and unsubstituted azetidin-2-ones to the corresponding anions is well known; anions as **2** have not been reported yet, as far as we know.

The organozinc **2** was easily prepared by stirring a *cis*/*trans* mixture of **1** with zinc powder in anhydrous THF at room temperature. The complete insertion of the metal was checked by quenching the reaction with water. The debrominated compound was quantitatively obtained af-

SYNTHESIS 2005, No. 1, pp 0061–0070
Advanced online publication: 02.11.2004
DOI: 10.1055/s-2004-834897; Art ID: Z15104SS

ter 30 minutes as a *cis*/*trans* mixture of *E*-alkenyl β-lactams.

The addition of benzaldehyde to the preformed enolate at room temperature did not afford any condensation product even after stirring the mixture overnight. In order to activate the electrophile, a solution of benzaldehyde/ $AlMe_2Cl$ in 1:1 ratio was added at 0 °C. The conversion of **2** into **3** and **4** proceeded within two hours at room temperature in good yield (Scheme 2).[9] A detectable amount of debrominated compound (about 15%) was also obtained. However, an improvement in the yield was observed under Barbier-type conditions, by directly mixing azetidinone **1**, zinc, the aldehyde, and one equivalent of $AlMe_2Cl$ in THF at room temperature.



Scheme 2 Generation of zinc-enolate **2** and reaction with aldehydes.

Detailed studies of the reaction showed a considerable influence of the solvent. THF and MeCN were found to be better solvents than other aprotic solvents. This Reformatsky reaction was tested with some aromatic and aliphatic aldehydes and the corresponding hydroxy derivatives **3** and **4** were always obtained in good yields and with complete *trans* diastereoselectivity at the ring substitution. The phenyl substituent at C-4 of the organozinc enolate blocks the approach of the electrophile on the more hindered face of the azetidin-2-one and a *trans* relationship between the newly introduced group and the phenyl substituent in C-4 was observed in the final product.[10] Unfortunately 1:1 mixtures of diastereomers **3** and **4** were obtained with aromatic aldehydes, while aliphatic aldehydes, such as isopropyl or cyclohexylcarbaldehyde gave increased yields and 70:30 diastereomeric ratios. The diastereomeric mixtures were easily separated by flash chromatography. Compounds **3a** and **3b**, with lower R_f, are solid products, which can be crystallized from EtOH. Their relative stereochemistry was assigned by X-ray crystallographic analysis (Figures 2 and 3).[11] The stereochemical assignments of the products **3c–e** and **4c–e,** were established on the basis of a detailed ^1H chemical shift comparison.

Further applications of this reaction were studied by reacting the substrates **1f–i** (Figure 4) with benzaldehyde (Table 1).

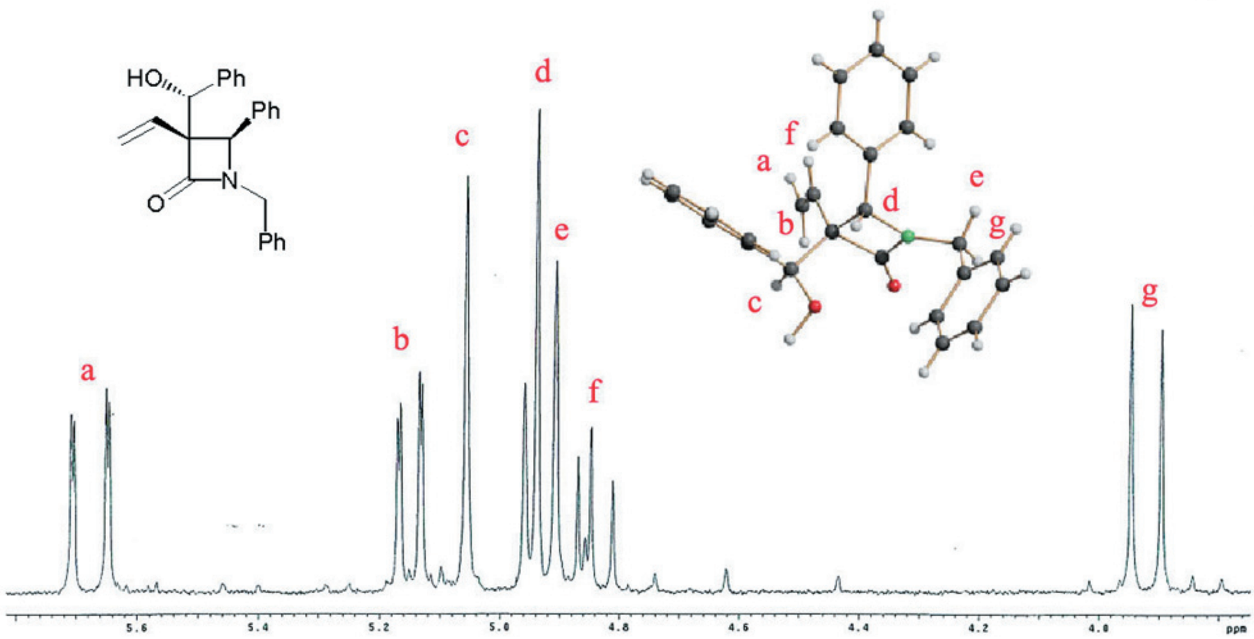


Figure 2 X-ray crystallographic analysis and assignments of the proton signals in the ^1H NMR spectra for **3a**.

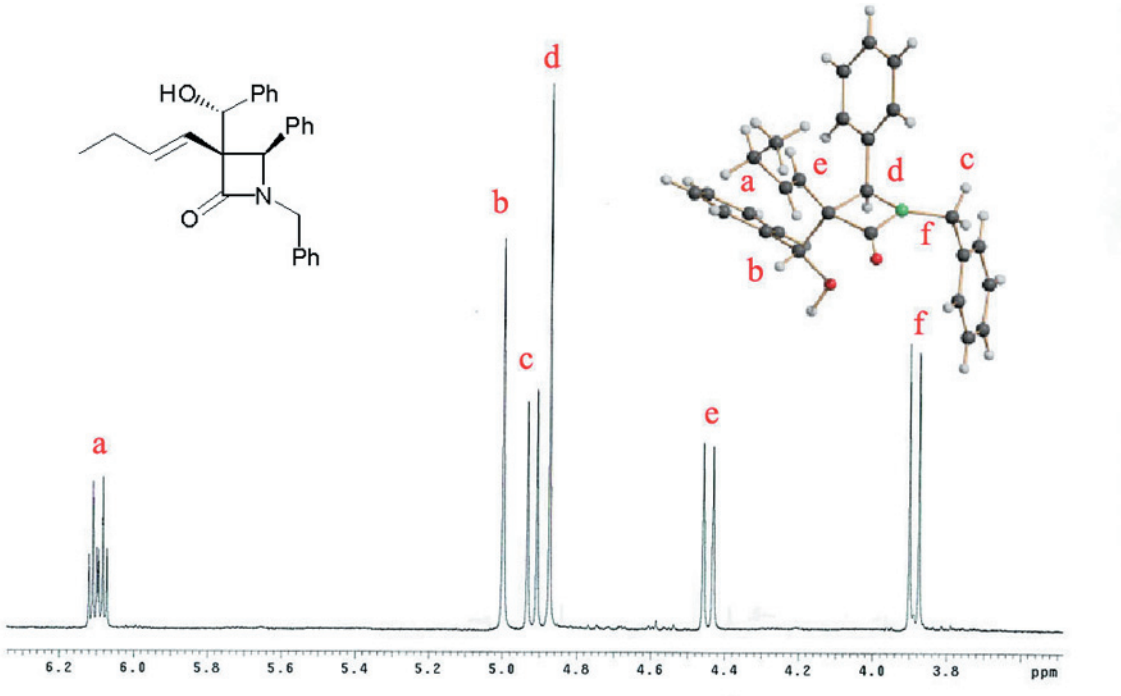


Figure 3 X-ray crystallographic analysis and assignments of the proton signals in the ^{1}H NMR spectra for **3b**.

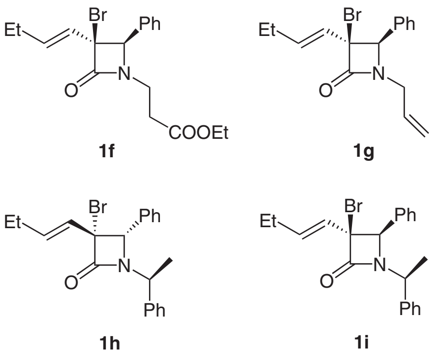


Figure 4 Azetidinones **1f–i**

Table 1 Reaction of Azetidinones **1f–i** with Benzaldehyde

Entry	Azetidinone	N-Substituent	Yield of **3** + **4** (%)
1	**1f**	$CH_2CH_2CO_2Et$	85
2	**1g**	$CH_2CH=CH_2$	91 (dr = 68:32)
3	**1h**	(*S*)-CH(CH_3)Ph	87
4	**1i**	(*S*)-CH(CH_3)Ph	72

The results reported show that the reaction is not influenced by the nature of the N-substituent, giving good yields for all the different starting azetidinones. When the reaction was performed on azetidinone **1g**, moderate diastereoselectivity was observed in the formation of hydroxyphenyl carbon centre (Table 1, entry 2). On the other hand, the introduction of a further stereogenic centre into the starting azetidinone (entries 3 and 4), did not change the stereochemical outcome of the reaction, affording derivatives **3h/4h** and **3i/4i** in an 1:1 diastereomeric ratio.

Lewis acids have been used to promote a variety of different reactions[12] and many attempts to classify them have been reported in the literature.[13] The reactivity of organozinc derivatives with aldehydes in the presence of metal salts has been recently reported in the literature. In order to establish the effect of Lewis acids on the reactivity of organozincs **2**, relatively oxophilic metals[13a,b] were investigated in the reaction of **2b** and benzaldehyde (Table 2).

Table 2 Lewis Acid Catalyzed Reaction of Azetidinones **1b** with Zinc and Benzaldehyde

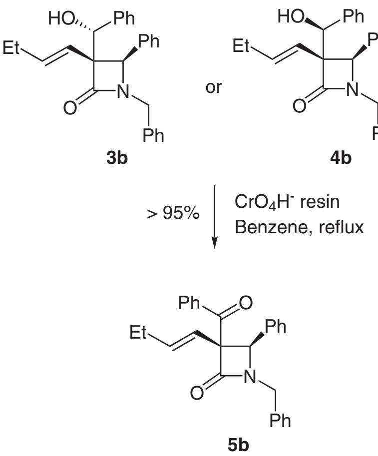
Entry	Lewis Acid (1 equiv)	Time (h)	Yield of **3b** + **4b** (%)
1	$Sc(OTf)_2$	24	–
2	$ZrCl_4$	6	30
3	$BF_3·OEt_2$	2	79
4	Ti(O*i*-Pr)$_4$	2	65
5	TMSCl	72	76
6	TMSCl + $AlMe_2Cl$ (10%)	2	80

While the addition of scandium triflate to the reaction mixture did not afford any product (Table 2, entry 1), the use of zirconium chloride allowed to obtain compounds

3b/4b in 30% yield (entry 2). By activating the reaction with $Ti(Oi\text{-}Pr)_4$ and $BF_3{\cdot}OEt_2$, good yields were observed within two hours (entries 3,4). Furthermore, the titanium-catalyzed reaction showed a moderate diastereoselectivity at the hydroxyphenyl carbon, affording compounds **3b/4b** in a 70:30 diastereomeric ratio.

Silicon Lewis acids[14] afford some advantages over traditional metal-centered activators. For example, they are compatible with many synthetically valuable carbon nucleophiles, such as silyl enol ethers, allyl organometallic reagents and cuprates. For these reasons, the Reformatsky reaction of **1b** was performed in the presence of 1 equivalent of trimethylsilyl chloride (entry 5). This condensation was very sluggish (72 h) and derivatives **3b/4b** were isolated in 76% yield, after treatment of the silyl intermediates with 6 N HCl during the workup (entry 5). Finally, good yield and a faster reaction were observed by adding 10% of $AlMe_2Cl$ to the TMSCl-catalyzed reaction (entry 6).

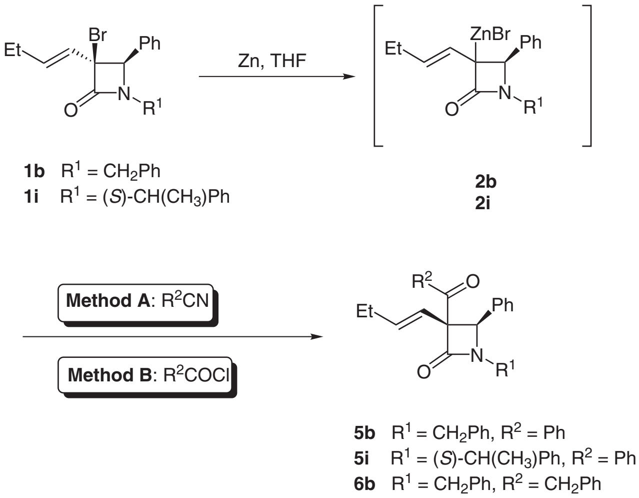
Oxidation of each diastereomeric hydroxyphenylazetidinone **3b** and **4b** with chromic acid on anion exchange resin,[15] furnished in each case a single keto derivative **5b**, in quantitative yield, thereby confirming that the hydroxyphenylazetidinones differed only in the stereochemistry of the side chain (Scheme 3).



Scheme 3 Oxidation of **3b** or **4b** to ketone **5b** with CrO_4H^- resin.

The oxidation was simply performed by stirring **3** or **4** in the presence of an excess of CrO_4H^- resin in refluxing benzene. The product was isolated by removing the resins by filtration and evaporating the solvent under reduced pressure.

In an effort to evaluate the ability of anion **2** to react in the Blaise reaction with nitriles,[16] the alternative synthesis of β,γ-unsaturated ketones was performed by treating **2b** with benzonitrile and phenylacetonitrile in THF under Barbier conditions (Scheme 4, Method A).



Scheme 4 Blaise reaction of **1b–i** with zinc and nitriles and reaction of organozinc **2** with acyl chlorides.

The reaction was sluggish (72 h), even in the presence of two equivalents of nitrile derivative (Table 3). Any attempt to improve the yield and to decrease the reaction time by adding a Lewis acid failed. Some selected results are reported in Table 3.

Table 3 Blaise Reaction of Azetidinones **1b** with Zinc and Nitriles

Entry	Azetidinone	R^2	Product	Yield (%)
1	**1b**	Ph	**5b**	50
2	**1b**	CH_2Ph	**5i**	60
5	**1i**	Ph	**6b**	60

In an alternative pathway (Scheme 4, Method B), anion **2b** was reacted with benzoyl chloride under the same conditions and ketone **5b** was obtained in 60% yield.

In summary, we have demonstrated that the in situ generated organozinc reagents, derived from 3-alkenyl-3-bromoazetidin-2-ones, react with aromatic and aliphatic aldehydes with complete regioselectivity and stereoselectivity on the ring substitution. The effect of solvent and the addition of Lewis acids have been studied in order to improve the reaction yields. The application of this methodology to nitriles and acyl chlorides also allowed the preparation of keto derivatives.

Unless stated otherwise, solvents and chemicals were obtained from commercial sources and used without further purification. Flash chromatography was performed on silica gel (230–400 mesh). NMR Spectra were recorded with 300 MHz or 600 MHz spectrometers. Chemical shifts were reported as δ values (ppm) relative to the solvent peak of $CDCl_3$ set at δ = 7.27 (^1H NMR) or δ = 77.0 (^{13}C NMR). IR spectra were recorded with an FT-IR spectrometer. Melting points are uncorrected. MS analyses were performed on a liquid chromatograph coupled with an electrospray ionization-mass spectrometer (LC-ESI-MS), using H_2O–MeCN as solvent at 25 °C (positive scan 100–500 *m*/*z*, fragmentor 70 eV).

3-Alkenyl-3-bromoazetidin-2-ones 1; General Procedure

To a refluxing solution of a respective imine (1 mmol) and Et_3N (2 mmol, 2 equiv, 0.280 mL) in CH_2Cl_2 (10 mL) under N_2, was added dropwise the corresponding acyl chloride (1.25 mmol, 1.25 equiv) in CH_2Cl_2 (2 mL) very slowly. The solution was refluxed for 4 h and then allowed to reach r.t. After quenching with 0.1 M HCl (5 mL), the mixture was diluted with CH_2Cl_2 (10 mL) and washed with H_2O (2 ×). The organic layer was dried (Na_2SO_4) and the solvent was removed under reduced pressure. The products were obtained, after purification by flash chromatography on silica gel (cyclohexane–Et_2O, 95:5 or benzene as eluent).

1-Benzyl-3-bromo-4-phenyl-3-vinylazetidin-2-one (1a)

Yield: 30%; *cis*/*trans* = 90:10; pale yellow oil.

^{1}H NMR ($CDCl_3$): δ = 3.92 (d, *J* = 14.8 Hz, 1 H, PhC*H*$_2$), 4.61 (br s, 1 H, CCHPh), 5.02 (d, *J* = 14.8 Hz, 1 H, PhC*H*$_2$), 5.35 (d, *J* = 10.7 Hz, 1 H, CH=C*H*$_2$), 5.65 (d, *J* = 16.8 Hz, 1 H, CH=C*H*$_2$), 6.18 (dd, *J* = 10.7, 16.8 Hz, 1 H, C*H*=CH$_2$), 7.1–7.5 (m, 10 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 44.7, 66.1, 69.5, 120.4, 127.6, 127.9, 128.2, 128.3, 128.7, 128.9, 131.8, 134.0, 134.3, 165.3.

LC-ESI-MS (R_t 12.2 min): *m*/*z* = 342/344 (M + 1), 364/366 (M + Na).

Anal. Calcd for $C_{18}H_{16}BrNO$: C, 63.17; H, 4.71; N, 4.09. Found: C, 63.15; H, 4.69; N, 4.11.

1-Benzyl-3-bromo-3-but-1-enyl-4-phenylazetidin-2-one (1b)

Yield: 60%; *cis*/*trans* = 84:16; pale yellow oil.

^{1}H NMR ($CDCl_3$): δ = 1.02 (t, *J* = 7.6 Hz, 3 H, CH$_2$C*H*$_3$), 2.13 (pseudo quintet, *J* = 6.4 Hz, 2 H, C*H*$_2$CH$_3$), 3.89 (d, *J* = 15.0 Hz, 1 H, PhC*H*$_2$), 4.55 (s, 1 H, CC*H*Ph), 4.98 (d, *J* = 15.0 Hz, 1 H, PhC*H*$_2$), 5.80 (dt, $J^{1,2}$ = 15.4 Hz, $J^{1,3}$ = 1.6 Hz, 1 H, C*H*=CHCH$_2$), 6.13 (dt, *J* = 15.4, 6.4 Hz, 1 H, CH=C*H*CH$_2$), 7.10–7.44 (m, 10 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 12.9, 25.4, 44.8, 66.9, 69.7, 125.5, 127.7, 127.9, 128.3, 128.4, 128.8, 128.9, 134.6, 136.8, 141.3, 165.7.

LC-ESI-MS (R_t 14.4 min): *m*/*z* = 370/372 (M + 1), 392/394 (M + Na).

Anal. Calcd for $C_{20}H_{20}BrNO$: C, 64.87; H, 5.44; N, 3.78. Found: C, 64.88; H, 5.43; N, 3.76.

3-(3-Bromo-3-but-1-enyl-2-oxo-4-phenylazetidin-1-yl)propionic Acid Ethyl Ester (1f)

Yield 55%; *cis*/*trans* = 93:7; pale yellow oil.

^{1}H NMR ($CDCl_3$): δ = 1.06 (t, *J* = 7.4 Hz, 3 H, CH$_2$C*H*$_3$), 1.22 (t, *J* = 7.5 Hz, 3 H, OCH$_2$C*H*$_3$), 2.17 (dq, *J* = 7.4, 6.3 Hz, 2 H, C*H*$_2$CH$_3$), 2.58 (dt, *J* = 16.8, 6.9 Hz, 1 H, C*H*$_2$CO$_2$Et), 2.73 (dt, *J* = 16.8, 6.9 Hz, 1 H, C*H*$_2$CO$_2$Et), 3.28 (dt, *J* = 14.4, 6.9 Hz, 1 H, NCH$_2$), 3.85 (dt, *J* = 14.4, 6.9 Hz, 1 H, NCH$_2$), 4.05 (q, *J* = 7.5 Hz, 2 H, OCH$_2$CH$_3$), 4.82 (br s, 1 H, CHPh), 5.85 (d, *J* = 15.3 Hz, 1 H, C*H*=CHCH$_2$), 6.14 (dt, *J* = 15.3, 6.3 Hz, 1 H, CH=C*H*CH$_2$), 7.21–7.45 (m, 5 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 11.3, 14.0, 25.2, 32.3, 46.6, 60.8, 64.7, 68.3, 125.9, 127.8, 128.5, 128.9, 136.7, 144.6, 165.6, 171.8.

LC-ESI-MS (R_t 13.40 min): *m*/*z* = 380/382 (M + 1), 402/404 (M + Na).

Anal. Calcd for $C_{18}H_{22}BrNO_3$: C, 56.85; H, 5.83; N, 3.68. Found: C, 56.84; H, 5.86; N, 3.68.

1-Allyl-3-bromo-3-but-1-enyl-4-phenylazetidin-2-one (1g)

Yield: 50%; *cis*/*trans* = 80:20; pale yellow oil.

^{1}H NMR ($CDCl_3$): δ = 1.07 (t, *J* = 7.8 Hz, 3 H, CH$_2$C*H*$_3$), 2.19 (ddq, $J^{1,2}$ = 6.2, 7.8 Hz, $J^{1,3}$ = 1.5 Hz, 2 H, C*H*$_2$CH$_3$), 3.48 (dd, *J* = 15.2, 7.0 Hz, 1 H, NC*H*$_2$), 4.31 (dd, *J* = 15.2, 5.2 Hz, 1 H, NC*H*$_2$), 4.81 (br s, 1 H, C*H*Ph), 5.08–5.23 (m, 2 H, CH=C*H*$_2$), 5.78 (m, 1 H, C*H*=CH$_2$), 5.91 (dt, $J^{1,2}$ = 15.4 Hz, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=CHCH$_2$), 6.20 (dt, *J* = 15.4, 6.2 Hz, 1 H, CH=C*H*CH$_2$), 7.23–7.28 (m, 2 H, C_6H_5), 7.38–7.45 (m, 3 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 12.8, 25.3, 43.2, 67.3, 69.6, 119.1, 127.8, 128.3, 128.9, 130.3, 134.7, 136.7, 144.6, 165.5.

LC-ESI-MS (R_t 12.40 min): *m*/*z* = 320/322 (M + 1), 342/344 (M + Na).

Anal. Calcd for $C_{16}H_{18}BrNO$: C, 60.01; H, 5.67; N, 4.37. Found: C, 60.00; H, 5.69; N, 4.34.

3-Bromo-3-but-1-enyl-4-phenyl-1-(1-phenylethyl)azetidin-2-ones

Yield: 55%; *cis*/*trans* >99/1; dr = 75:25

(1*S′*,3*R*,4*S*)-1h

Pale yellow oil; $[\alpha]_D$ +3 (*c* = 1, $CHCl_3$).

^{1}H NMR ($CDCl_3$): δ =1.02 (t, *J* = 7.5 Hz, 3 H, CH$_2$C*H*$_3$), 1.50 (d, *J* = 7.2 Hz, 3 H, CHC*H*$_3$), 2.05–2.22 (m, 2 H, C*H*$_2$CH$_3$), 4.50 (s, 1 H, CC*H*Ph), 5.10 (q, *J* = 7.2 Hz, 1 H, C*H*CH$_3$), 5.74 (dt, $J^{1,2}$ = 15.4 Hz, $J^{1,3}$ = 1.2 Hz, 1 H, C*H*=CHCH$_2$), 6.03 (dt, *J* = 15.4, 6.6 Hz, 1 H, CH=C*H*CH$_2$), 7.25–7.42 (m, 10 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.0, 18.8, 25.4, 53.1, 67.5, 68.8, 125.4, 127.3, 128.0, 128.3, 128.4, 128.7, 129.0, 136.0, 136.5, 139.3, 166.2.

LC-ESI-MS (R_t 15.57 min): *m*/*z* = 384/386 (M + 1), 406/408 (M + Na).

Anal. Calcd for $C_{21}H_{22}BrNO$: C, 65.63; H, 5.77; N, 3.64. Found: C, 65.60; H, 5.78; N, 3.66.

(1*S′*,3*S*,4*R*)-1i

White solid, mp 83–85 °C; $[\alpha]_D$ –10 (*c* = 1.2, $CHCl_3$).

^{1}H NMR ($CDCl_3$): δ = 1.07 (t, *J* = 7.5 Hz, 3 H, CH$_2$C*H*$_3$), 1.95 (d, *J* = 7.0 Hz, 3 H, CHC*H*$_3$), 2.10–2.22 (m, 2 H, C*H*$_2$CH$_3$), 4.33 (q, *J* = 7.0 Hz, 1 H, C*H*CH$_3$), 4.53 (s, 1 H), 5.83 (dt, $J^{1,2}$ = 15.4 Hz, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=CHCH$_2$), 6.18 (dt, *J* = 15.4, 4.4 Hz, 1 H, CH=C*H*CH$_2$), 7.22–7.43 (m, 10 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ =13.0, 20.3, 25.3, 55.4, 67.0, 69.1, 125.8, 126.7, 127.9, 128.0, 128.5, 128.8, 128.9, 135.0, 136.7, 140.8, 165.8.

LC-ESI-MS (R_t 15.76 min): *m*/*z* = 384/386 (M + 1), 406/408 (M + Na).

Anal. Calcd for $C_{21}H_{22}BrNO$: C, 65.63; H, 5.77; N, 3.64. Found: C, 65.61; H, 5.75; N, 3.67.

Reformatski-Type Reaction on 1; General Procedure

Method A: To a stirred solution of **1** (1 mmol) in anhyd THF (5 mL), under N_2 at r.t., was added zinc powder (1.5 mmol, 1.5 equiv, 65 mg.) in one portion. After 20 min, the temperature was lowered to 0 °C and a solution of benzaldehyde (1.5 mmol, 1.5 equiv, 0.15 mL) and $AlMe_2Cl$ (1.5 mmol, 1.5 equiv, 1.5 mL 1 M solution in hexane) in anhyd THF (3 mL) was added dropwise under inert atmosphere by means of a Teflon cannula. The reaction was monitored by TLC and stopped after 2 h by adding an aq sat. solution of Seignette salts (sodium potassium tartrate, 2 mL). The mixture was then filtered on a Celite pad and the organic solvent was concentrated under reduced pressure. The residue, diluted with CH_2Cl_2 (10 mL) was washed with H_2O (2 ×). The organic layer was dried (Na_2SO_4) and the solvent was removed. Compounds **3** and **4** were isolated by flash chromatography on silica gel (cyclohexane–EtOAc, 9:1 as eluent).

Method B: To a stirred solution of **1** (1 mmol) in anhyd THF (5 mL), were added zinc powder (1.5 mmol, 1.5 equiv, 65 mg.), benzaldehyde (1.5 mmol, 1.5 equiv., 0.15 mL) and $AlMe_2Cl$ (1.5 mmol, 1.5 equiv, 1.5 mL 1 M solution in hexane) were added under N_2 at r.t.

The reaction was monitored by TLC and stopped after 2 h by adding an aq sat. solution of Seignette salts (sodium potassium tartrate, 2 mL). The mixture was then filtered on a Celite pad and the organic solvent was concentrated under reduced pressure. The residue, diluted with CH_2Cl_2 (10 mL) was washed with H_2O (2 ×). The organic layer was dried (Na_2SO_4) and the solvent was removed. Compounds **3** and **4** were isolated by flash chromatography on silica gel (cyclohexane–EtOAc, 9:1 as eluent).

1-Benzyl-3-(hydroxyphenylmethyl)-4-phenyl-3-vinylazetidin-2-ones

3a

White solid; yield: 64%; mp 158–160 °C; dr = 50:50.

IR (Nujol): 3383, 2919, 2853, 1726, 1474, 1375, 1049, 1023 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 3.45 (d, *J* = 3.8 Hz, 1 H, OH), 3.98 (d, *J* = 15.0 Hz, 1 H, PhC*H*$_2$), 4.88 (dd, *J* = 11.3, 17.5 Hz, 1 H, C*H*=CH$_2$), 4.96 (d, *J* = 15.0 Hz, 1 H, PhC*H*$_2$), 4.98 (s, 1 H, C*H*Ph), 5.05 (s, 1 H, C*H*OH), 5.17 (dd, *J* = 1.5, 11.3 Hz, 1 H, CH=C*H*$_2$), 5.70 (dd, *J* = 1.5, 17.5 Hz, 1 H, CH=C*H*$_2$), 6.90–6.98 (m, 2 H, C_6H_5), 7.2–7.5 (m, 13 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 44.4, 59.4, 71.8, 75.5, 119.2, 127.5, 127.8, 128.1, 128.2, 128.3, 128.4, 128.5, 128.7, 128.8, 132.7, 135.1, 135.5, 140.0, 170.2.

LC-ESI-MS (R_t 13.19 min): *m/z* = 352 (M – H_2O + 1), 370 (M + 1), 392 (M + Na).

Anal. Calcd for $C_{25}H_{23}NO_2$: C, 81.27; H, 6.27; N, 3.79. Found: C, 81.26; H, 6.27; N, 3.81.

Single Crystal X-ray Diffraction Data of 3a

MF $C_{25}H_{23}NO_2$, Fw 369.44, colorless parallelepiped, size: 0.30 × 0.23 × 0.20 mm, monoclinic, space group $P2_1/n$, *a* = 12.4900(7) Å, *b* = 10.7157(5) Å, *c* = 15.5174(8) Å, β = 92.288(2)°, V = 2075.18(19) Å^3, theta range for data collection is from 2.05 to 30.00°, T = 293(2) K, *Z* = 4, *F*(000) = 784, D_x = 1.183 Mg/m^3, μ = 0.074 mm^{-1}, data collected on a Bruker AXS CCD diffractometer (Mo-Kα radiation, λ = 0.71073 Å) at 293(2) K, total of 23637 reflections, of which 6053 unique [$R_{(int)}$ = 0.1147]; 2191reflections $I > 2\sigma$ (I). Empirical absorption correction was applied, initial structure model by direct methods. Anisotropic full-matrix least-squares refinement on F^2 for all non-hydrogen atoms yielded $R1$ = 0.0606 and $wR2$ = 0.1382 for 2191 [$I > 2\sigma$ (I)] and $R1$ = 0.1824 and $wR2$ = 0.1812 for all (6053) intensity data. Goodness-of-fit = 0.851, the max/mean shift/esd is 0.00 and 0.00, max/min, CCDC number is 245012.

4a

Pale yellow oil.

IR (neat): 3417, 3069, 3022, 2922, 2845, 1731, 1649, 1560, 1490, 1460, 1401 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 2.64 (d, *J* = 2.1 Hz, 1 H, OH), 3.85 (d, *J* = 15.0 Hz, 1 H, PhC*H*$_2$), 4.65 (s, 1 H, C*H*Ph), 4.88 (d, 1 H, *J* = 15.0 Hz, PhC*H*$_2$), 5.13 (d, *J* = 2.1 Hz, 1 H, C*H*OH), 5.15–5.22 (m, 2 H, C*H*=C*H*$_2$), 5.59–5.67 (m, 1 H, CH=C*H*$_2$), 6.80–6.90 (m, 2 H, C_6H_5), 7.00–7.10 (m, 2 H, C_6H_5), 7.20–7.40 (m, 9 H, C_6H_5), 7.50–7.60 (m, 2 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 43.9, 60.3, 70.9, 76.3, 119.5, 127.4, 127.5, 127.8, 128.0, 128.2, 128.3, 128.4, 128.5, 128.6, 131.2, 134.6, 134.8, 139.1, 169.2.

LC-ESI-MS (R_t 12.90 min): *m/z* = 352 (M – H_2O + 1), 392 (M + Na).

Anal. Calcd for $C_{25}H_{23}NO_2$: C, 81.27; H, 6.27; N, 3.79. Found: C, 81.28; H, 6.25; N, 3.80.

1-Benzyl-3-but-1-enyl-3-(hydroxyphenylmethyl)-4-phenylazetidin-2-ones

3b

Yield: 64%; white solid; mp 159–161 °C; dr = 50:50.

IR (Nujol): 3364, 2959, 2933, 2853, 1733, 1461, 1375, 1063 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.81 (t, *J* = 7.5 Hz, 3 H, CH$_2$C*H*$_3$), 1.8–2.0 (m, 2 H, C*H*$_2$CH$_3$), 3.39 (d, *J* = 1.8 Hz, 1 H, OH), 3.90 (d, *J* = 15.3 Hz, 1 H, PhC*H*$_2$), 4.45 (d, *J* = 15.9 Hz, 1 H, C*H*=CHCH$_2$), 4.88 (s, 1 H, C*H*Ph); 4.92 (d, *J* = 15.3 Hz, 1, PhC*H*$_2$), 5.00 (s, 1 H, C*H*OH), 6.13 (dt, *J* = 15.9, 6.6 Hz, 1 H, CH=C*H*CH$_2$), 6.9–7.0 (m, 2 H, C_6H_5), 7.20–7.60 (m, 13 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.2, 25.6, 43.9, 59.3, 70.7, 75.4, 123.1, 127.2, 127.3, 127.6, 127.8, 127.9, 128.0, 128.3, 128.4, 134.9, 135.2, 136.4, 139.8, 170.2.

LC-ESI-MS (R_t 14.63 min): *m/z* = 379 (M – H_2O + 1), 398 (M + 1), 420 (M + Na).

Anal. Calcd for $C_{27}H_{27}NO_2$: C, 81.58; H, 6.85; N, 3.52. Found: C, 81.59; H, 6.87; N, 3.51.

Single Crystal X-ray Diffraction Data of 3b

MF $C_{27}H_{27}NO_2$, Fw: 397.50, colorless parallelepiped, size: 0.35 × 0.25 × 0.20 mm, triclinic, space group *P* –1, *a* = 10.134(3) Å, *b* = 12.498(3) Å, *c* = 19.626(5) Å, α = 82.360(6)° β = 87.201(6)°, γ = 68.077(6)°, V = 2285.5(10) Å^3, theta range for data collection is from 1.05 to 30.11°, T = 293(2) K, *Z* = 4, *F*(000) = 848, D_x = 1.155 Mg/m^3, μ = 0.072 mm^{-1}, data collected on a Bruker AXS CCD diffractometer (Mo-Kα radiation, λ = 0.71073 Å) at 293(2) K, total of 29731 reflections, of which 13315 unique [$R_{(int)}$ = 0.0843]; 3400 reflections $I > 2\sigma$ (I). Empirical absorption correction was applied, initial structure model by direct methods. Anisotropic full-matrix least-squares refinement on F^2 for all non-hydrogen atoms yielded $R1$ = 0.0653 and $wR2$ = 0.1465 for 3400 [$I > 2\sigma$ (I)] and $R1$ = 0.2529 and $wR2$ = 0.1994 for all (13315) intensity data. Goodness-of-fit = 0.784, the max/mean shift/esd is 0.00 and 0.00, max/min. CCDC number is 245013.

4b

Yellow oil.

IR (neat): 3423, 2965, 2913, 1739, 1653, 1448, 1408, 1355 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.83 (t, *J* = 7.5 Hz, 3 H, CH$_2$C*H*$_3$), 1.9–2.0 (m, 2 H, C*H*$_2$CH$_3$), 3.39 (d, *J* = 1.5 Hz, 1 H, OH), 3.82 (d, *J* = 15.3 Hz, 1 H, PhC*H*$_2$), 4.64 (s, 1 H, C*H*Ph), 4.78 (dt, $J^{1,2}$ = 15.8, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=CHCH$_2$), 4.88 (d, *J* = 15.3 Hz, 1 H, PhC*H*$_2$), 5.08 (d, *J* = 1.5 Hz, 1 H, C*H*OH), 6.04 (dt, *J* = 6.6, 15.8 Hz, 1 H, CH=C*H*CH$_2$), 6.8–7.6 (m, 15 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.2, 26.8, 43.7, 60.4, 70.1, 76.3, 121.9, 127.3, 127.4, 127.7, 127.8, 127.9, 128.1, 128.5, 134.8, 134.9, 136.9, 139.4, 169.7.

LC-ESI-MS (R_t 14.17 min): *m/z* = 379 (M – H_2O + 1), 420 (M + Na).

Anal. Calcd for $C_{27}H_{27}NO_2$: C, 81.58; H, 6.85; N, 3.52. Found: C, 81.61; H, 6.85; N, 3.49.

1-Benzyl-3-but-1-enyl-3-[hydroxy(4-methoxyphenyl)methyl]-4-phenylazetidin-2-ones

3c

Yield: 64%; yellow waxy oil; dr = 50:50.

IR (neat): 3364, 3025, 3006, 2952, 1726, 1613, 1507, 1447, 1242, 1169, 1030 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.81 (t, *J* = 7.2 Hz, 3 H, CH$_2$C*H*$_3$); 1.81–1.96 (m, 2 H, C*H*$_2$CH$_3$), 2.57 (br s, 1 H, OH), 3.83 (s, 3 H, OCH$_3$), 3.88 (d, *J* = 15.3 Hz, 1 H, PhC*H*$_2$), 4.48 (d, *J* = 15.4 Hz, 1 H, C*H*=CHCH$_2$), 4.84 (s, 1 H, C*H*Ph), 4.93 (d, J = 15.3 Hz, 1 H, PhC*H$_2$*), 4.94 (br s, 1 H, C*H*OH), 6.10 (dt, J = 15.4, 6.6 Hz, 1 H, CH=C*H*CH$_2$), 6.80–6.90 (m, 2 H, C_6H_5), 6.91–7.00 (m, 1 H, C_6H_5), 7.01–7.10 (m, 1 H, C_6H_5), 7.20–7.40 (m, 10 H, C_6H_5).

^{13}C NMR (CDCl$_3$): δ = 13.2, 25.7, 43.8, 55.2, 59.5, 71.0, 75.2, 113.5, 123.1, 127.4, 127.8, 128.1, 128.2, 128.3, 128.4, 128.5, 131.7, 134.9, 135.2, 136.8, 159.3, 168.9.

LC-ESI-MS (R$_t$ 15.20 min): m/z = 410 (M – H$_2$O + 1), 428 (M + 1), 450 (M + Na), 877 (2 M + Na).

Anal. Calcd for $C_{28}H_{29}NO_2$: C, 81.72; H, 7.10; N, 3.40. Found: C, 81.71; H, 7.08; N, 3.41.

4c
Yellow oil.

IR (neat): 3430, 3025, 2966, 2926, 1726, 1600, 1514, 1454, 1235, 1169, 1036 cm^{-1}.

^{1}H NMR (CDCl$_3$): δ = 0.79 (t, J = 7.3 Hz, 3 H, CH$_2$C*H$_3$*), 1.81–1.96 (m, 2 H, CH$_2$C*H$_3$*), 2.61 (br s, 1 H, OH), 3.78 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 3.84 (s, 3 H, OC*H$_3$*), 4.59 (s, 1 H, C*H*Ph), 4.77 (dt, $J^{1,2}$ = 15.8 Hz, $J^{1,3}$ = 1.4 Hz, 1 H, C*H*=CHCH$_2$), 4.86 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 5.02 (s, 1 H, C*H*OH), 6.00 (dt, J = 15.8, 6.6 Hz, 1 H, C*H*=CHCH$_2$), 6.80–6.90 (m, 4 H, C_6H_5), 6.91–7.10 (m, 2 H, C_6H_5), 7.11–7.30 (m, 6 H, C_6H_5), 7.31–7.40 (m, 2 H, C_6H_5).

^{13}C NMR (CDCl$_3$): δ = 13.2, 25.8, 43.7, 55.2, 60.3, 70.4, 76.0, 113.4, 122.1, 127.4, 127.7, 127.8, 127.9, 128.2, 128.5, 128.6, 131.5, 134.8, 134.9, 136.9, 159.3, 169.7.

LC-ESI-MS (R$_t$ 15.08 min): m/z = 410 (M – H$_2$O + 1), 428 (M + 1), 450 (M + Na), 877 (2 M + Na).

Anal. Calcd for $C_{28}H_{29}NO_2$: C, 81.72; H, 7.10; N, 3.40. Found: C, 81.74; H, 7.11; N, 3.38.

1-Benzyl-3-but-1-enyl-3-(1-hydroxy-2-methylpropyl)-4-phenylazetidin-2-ones
3d
Yield: 86%; colorless oil; dr = 70:30.

IR (neat): 3450, 2966, 2926, 2739, 1454, 1394, 1348, 1149 cm^{-1}.

^{1}H NMR (CDCl$_3$): δ = 0.90 (t, J = 7.3 Hz, 3 H, CH$_2$C*H$_3$*), 0.91 (d, J = 6.0 Hz, 3 H, C*H$_3$*CHCH$_3$), 1.00 (d, J = 6.0 Hz, 3,H, CH$_3$CHC*H$_3$*), 1.8–2.0 (m, 3 H, C*H$_2$*CH$_3$ + CH$_3$C*H*CH$_3$), 2.20 (br s, 1 H, OH), 3.85 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 3.91 (d, J = 3.6 Hz, 1 H, C*H*OH), 4.81 (dt, $J^{1,2}$ = 15.9 Hz, $J^{1,3}$ = 1.4 Hz, 1 H, C*H*=CHCH$_2$), 4.85 (s, 1 H, C*H*Ph), 4.87 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 6.10 (dt, J = 15.9, 6.6 Hz, 1 H, CH=C*H*CH$_2$), 7.10–7.50 (m, 10 H, C_6H_5).

^{13}C NMR (CDCl$_3$): δ = 13.3, 16.5, 20.5, 25.8, 29.4, 43.8, 60.2, 70.2, 77.8, 122.6, 127.4, 128.0, 128.2, 128.5, 128.6, 129.3, 134.9, 135.6, 135.7, 170.9.

LC-ESI-MS (R$_t$ 17.20 min): m/z = 346 (M – H$_2$O + 1), 364 (M + 1), 386 (M + Na), 402 (M + K).

Anal. Calcd for $C_{24}H_{29}NO_2$: C, 79.30; H, 8.04; N, 3.85. Found: C, 79.31; H, 8.01; N, 3.86.

4d
White solid; mp 85–87 °C.

IR (Nujol): 3463, 3032, 2966, 2933, 2873, 1752, 1494, 1454, 1388, 1355, 1063 cm^{-1}.

^{1}H NMR (CDCl$_3$): δ = 0.88 (t, J = 7.2 Hz, 3 H, CH$_2$C*H$_3$*), 0.89 (d, J = 6.6 Hz, 3 H, C*H$_3$*CHCH$_3$), 1.05 (d, J = 6.6 Hz, 3 H, CH$_3$CHC*H$_3$*), 1.6–1.8 (m, 1 H, CH$_3$C*H*CH$_3$), 1.85–2.00 (m, 2 H, C*H$_2$*CH$_3$), 2.84 (br s, 1 H, OH), 3.60 (d, J = 7.0 Hz, 1 H, C*H*OH), 3.79 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 4.38 (s, 1 H, C*H*Ph), 4.73 (dt, $J^{1,2}$ = 15.6 Hz, $J^{1,3}$ = 1.4 Hz, 1 H, C*H*=CHCH$_2$), 4.94 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 6.09 (dt, J = 15.6, 6.6 Hz, 1 H, CH=C*H*CH$_2$), 7.00–7.20 (m, 4 H, C_6H_5), 7.21–7.5 (m, 6 H, C_6H_5).

^{13}C NMR (CDCl$_3$): δ = 13.2, 18.3, 19.7, 25.9, 30.8, 43.7, 63.0, 69.8, 80.0, 120.6, 127.5.127.7, 128.0, 128.1, 128.4, 128.5, 128.7, 129.0, 134.1, 135.5, 137.1, 168.9.

LC-ESI-MS (R$_t$ 16.30 min): m/z = 346 (M - H$_2$O + 1), 364 (M + 1), 386 (M + Na), 402 (M + K).

Anal. Calcd for $C_{24}H_{29}NO_2$: C, 79.30; H, 8.04; N, 3.85. Found: C, 79.30; H, 8.02; N, 3.84.

1-Benzyl-3-but-1-enyl-3-(cyclohexylhydroxymethyl)-4-phenylazetidin-2-ones
3e
Yield: 67%; colorless oil; dr = 70:30.

IR (neat): 3436, 2959, 2926, 2853, 1739, 1666, 1441, 1387, 1341, 1069 cm^{-1}.

^{1}H NMR (CDCl$_3$): δ = 0.88 (t, J = 7.5 Hz, 3 H, CH$_2$C*H$_3$*), 1.00–1.10 (m, 2 H, CH$_2$, cyclohexyl), 1.20–1.30 (m, 4 H, CH$_2$, cyclohexyl), 1.40–1.50 (m, 2 H, CH$_2$, cyclohexyl), 1.56 (m, 1 H, CH, cyclohexyl), 1.60–1.70 (m, 2 H, CH$_2$, cyclohexyl), 1.90–2.00 (m, 2 H, C*H$_2$*CH$_3$), 3.48 (br s, 1 H, OH), 3.80 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 3.85 (br s, 1 H, C*H*OH), 4.77 (s, 1 H, C*H*Ph), 4.78 (d, J = 15.8 Hz, 1 H, C*H*=CHCH$_2$), 4.84 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 6.05 (dt, J = 15.8, 6.6 Hz, 1 H, CH=C*H*CH$_2$), 7.10–7.40 (m, 10 H, C_6H_5).

^{13}C NMR (CDCl$_3$): δ = 13.4, 25.9, 26.0, 26.4, 26.9, 30.4, 30.6, 39.5, 43.8, 60.4, 70.1, 77.9, 122.5, 127.4, 127.9, 128.1, 128.2, 128.4, 128.5, 128.7, 129.3, 134.5, 134.9, 135.8, 170.8.

LC-ESI-MS (R$_t$ 19.40 min): m/z = 386 (M – H$_2$O + 1), 404 (M + 1), 426 (M + Na), 442 (M + K).

Anal. Calcd for $C_{27}H_{33}NO_2$: C, 80.36; H, 8.24; N, 3.47. Found: C, 80.35; H, 8.22; N, 3.50.

4e
White solid; mp 100–102 °C.

IR (Nujol): 3463, 3025, 2959, 2926, 2853, 1759, 1493, 1454, 1387, 1398, 1361, 1069 cm^{-1}.

^{1}H NMR (CDCl$_3$): δ = 0.91 (t, J = 7.5 Hz, 3 H, CH$_2$C*H$_3$*), 1.05–1.10 (m, 4 H, CH$_2$, cyclohexyl), 1.25–1.35 (m, 2 H, CH$_2$, cyclohexyl), 1.60–1.80 (m, 4 H, CH$_2$, cyclohexyl), 1.90–2.10 (m, 2 H, C*H$_2$*CH$_3$), 2.05–2.10 (m, 1 H, CH, cyclohexyl), 2.90 (br s, 1 H, OH), 3.64 (d, J = 6.9 Hz, 1 H, C*H*OH), 3.78 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 4.37 (s, 1 H, C*H*Ph), 4.75 (d, J = 15.6 Hz, 1 H, C*H*=CHCH$_2$), 4.91 (d, J = 15.0 Hz, 1 H, PhC*H$_2$*), 6.11 (dt, 1 H, J = 15.6, 6.6 Hz, CH=C*H*CH$_2$), 7.10–7.20 (m, 4 H, C_6H_5), 7.30–7.40 (m, 6 H, C_6H_5).

^{13}C NMR (CDCl$_3$): δ = 13.3, 25.8, 25.9, 26.1, 26.2, 28.6, 29.6, 40.5, 43.7, 62.9, 69.6, 79.3, 120.6, 127.7, 128.1, 128.3, 128.4, 128.7, 129.0, 134.1, 135.3, 137.2, 170.4.

LC-ESI-MS (R$_t$ 19.50 min): m/z = 386 (M – H$_2$O + 1), 404 (M + 1), 426 (M + Na).

Anal. Calcd for $C_{27}H_{33}NO_2$: C, 80.36; H, 8.24; N, 3.47. Found: C, 80.38; H, 8.22; N, 3.49.

3-[3-But-1-enyl-3-(hydroxyphenylmethyl)-2-oxo-4-phenylazetidin-1-yl]propionic Acid Ethyl Esters
3f
Yield: 85%; white solid; mp 89–91 °C; dr = 50:50.

IR (neat): 3430, 3065, 3032, 2972, 2933, 1746, 1494, 1441, 1401, 1375, 1189, 1050, 1017 cm^{-1}.

^{1}H NMR (CDCl$_3$): δ = 0.79 (t, J = 7.5 Hz, 3 H, CH$_2$C*H$_3$*), 1.25 (t, J = 6.9 Hz, 3 H, OCH$_2$C*H$_3$*), 1.8–1.9 (m, 2 H, C*H$_2$*CH$_3$), 2.58 (br t, J = 6.9 Hz, 2 H, C*H$_2$*CO$_2$Et), 3.23 (dt, J = 14.1, 6.9 Hz, 1 H, NC*H$_2$*), 3.54 (br s, 1 H, OH), 3.84 (dt, J = 14.1, 6.9 Hz, 1 H, NC*H$_2$*), 4.05–

4.15 (m, 2 H, OCH_2CH_3), 4.39 (dt, $J^{1,2}$ = 15.9 Hz, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=CH-CH_2), 5.03 (s, 1 H, C*H*OH), 5.13 (br s, 1 H, C*H*Ph), 6.03 (dt, *J* = 15.9, 6.6 Hz, 1 H, CH=C*H*CH_2), 6.96–6.99 (m, 2 H, C_6H_5), 7.20–7.50 (m, 8 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.1, 14.0, 25.6, 32.6, 36.1, 59.7, 60.7, 70.6, 75.2, 122.8, 127.1, 127.6, 127.8, 127.9, 128.0, 128.2, 135.3, 136.2, 139.8, 170.6, 171.2.

LC-ESI-MS (R_t 13.40 min): *m*/*z* = 390 (M – H_2O + 1), 408 (M + 1), 430 (M + Na).

Anal. Calcd for $C_{25}H_{29}NO_4$: C, 73.68; H, 7.17; N, 3.44. Found: C, 73.66; H, 7.19; N, 3.44.

4f
Yellow oil.

IR (neat): 3423, 3059, 3032, 2966, 2926, 1726, 1501, 1441, 1375, 1189, 1056 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.77 (t, 3 H, *J* = 7.5 Hz, CH_2CH_3), 1.24 (t, *J* = 7.0 Hz, 3 H, OCH_2CH_3), 1.8–1.9 (m, 2 H, CH_2CH_3), 2.28 (br t, *J* = 6.9 Hz, 2 H, CH_2CO_2Et), 2.81 (br s, 1 H, OH), 3.11 (dt, *J* = 14.1, 6.9 Hz, 1 H, NCH_2), 3.76 (dt, *J* = 14.1, 6.9 Hz, 1 H, NCH_2), 4.00–4.20 (m, 2 H, OCH_2CH_3), 4.71 (dt, $J^{1,2}$ = 15.9 Hz, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=CHCH_2), 4.85 (s, 1 H, C*H*Ph), 5.11 (br s, 1 H, C*H*OH), 5.95 (dt, *J* = 15.9, 6.6 Hz, 1 H, CH=C*H*CH_2), 7.00–7.10 (m, 2 H, C_6H_5), 7.20–7.60 (m, 8 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.1, 14.0, 25.7, 32.4, 35.7, 60.7, 61.5, 70.2, 76.4, 121.4, 126.9, 127.0, 127.4, 127.7, 127.9, 128.0, 128.2, 128.4, 135.2, 137.1, 139.4, 169.9, 171.0.

LC-ESI-MS (R_t 13.06 min): *m*/*z* = 390 (M – H_2O + 1), 408 (M + 1), 430 (M + Na).

Anal. Calcd for $C_{25}H_{29}NO_4$: C, 73.68; H, 7.17; N, 3.44. Found: C, 73.67; H, 7.17; N, 3.47.

1-Allyl-3-but-1-enyl-3-(hydroxyphenylmethyl)-4-phenylazetidin-2-ones
3g
Yield: 91%; colorless oil; dr = 68:32.

IR (neat): 3403 3025 2952 2926 1733 1494 1454 1401 1355 1288 1050 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.81 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.8–2.0 (m, 2 H, CH_2CH_3), 2.76 (br s, 1 H, OH), 3.44 (dd, *J* = 15.6, 6.9 Hz, 1 H, NCH_2), 4.28 (dd, *J* = 15.6, 4.8 Hz, 1 H, NCH_2), 4.45 (d, *J* = 15.9 Hz, 1 H, C*H*=CHCH_2), 5.07 (s, 1 H, C*H*Ph), 5.13–5.22 (m, 2 H, CH=CH_2), 5.16 (s, 1 H, C*H*OH), 5.60–5.80 (m, 1 H, C*H*=CH_2), 6.08 (dt, *J* = 15.6, 6.6 Hz, 1 H, CH=C*H*CH_2), 6.90–7.00 (m, 2 H, C_6H_5), 7.20–7.50 (m, 8 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.2, 25.6, 42.7, 59.2, 70.6, 75.2, 118.4, 123.0, 127.2, 127.6, 127.8, 127.9, 128.0, 128.1, 128.2,128.3, 131.5, 135.3, 136.3, 140.0, 170.3.

LC-ESI-MS (R_t 13.52 min): *m*/*z* = 330 (M – H_2O + 1), 370 (M + Na).

Anal. Calcd for $C_{23}H_{25}NO_2$: C, 79.51; H, 7.25; N, 4.03. Found: C, 79.52; H, 7.28; N, 4.02.

4g
Colorless oil.

IR (neat): 3417 3063 3034 2969 2921 1725 1696 1649 1560 1448 1401 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.78 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.8–2.0 (m, 2 H, CH_2CH_3), 2.60 (br s, 1 H, OH), 3.33 (dd, *J* = 15.3, 7.2 Hz, 1 H, NCH_2), 4.20 (dd, *J* = 15.3, 4.8 Hz, 1 H, NCH_2), 4.72 (dt, $J^{1,2}$ = 15.6 Hz, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=CHCH_2), 4.81 (s, 1 H, C*H*Ph), 4.93–5.06 (m, 2 H, CH=CH_2), 5.14 (s, 1 H, C*H*OH), 5.3–5.5 (m, 1 H, C*H*=CH_2), 5.97 (dt, *J* = 15.6, 6.6 Hz, 1 H, CH=C*H*CH_2), 7.06 (d, *J* = 7.8 Hz, 2 H, C_6H_5), 7.2–7.5 (m, 6 H, C_6H_5), 7.57 (d, *J* = 7.8 Hz, 2 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.2, 26.8, 42.6, 60.6, 72.4, 74.2, 118.6, 122.9, 127.2, 127.6, 127.8, 127.9, 128.0, 128.0, 128.1, 128.2, 131.6, 135.6, 136.6, 139.8, 170.3.

LC-ESI-MS (R_t 13.34 min): *m*/*z* = 330 (M – H_2O + 1), 370 (M + Na).

Anal. Calcd for $C_{23}H_{25}NO_2$: C, 79.51; H, 7.25; N, 4.03. Found: C, 79.50; H, 7.26; N, 4.03.

3-But-1″-enyl-3-(hydroxyphenylmethyl)-4-phenyl-1-(1′-phenylethyl)azetidin-2-ones
(1′*S*,1″*R*,3*R*,4*S*)-3h
Yellow oil; yield: 87%, dr = 50:50; $[\alpha]_D$ +85 (*c* = 1.1, $CHCl_3$).

IR (neat): 3396, 3064, 3031, 2958, 2932, 1725, 1493, 1453, 1340, 1201, 1055, 1022 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.83 (t, *J* = 7.2 Hz, 3 H, CH_2CH_3), 1.50 (d, *J* = 7.2 Hz, 3 H, $CHCH_3$), 1.8–2.0 (m, 2 H, CH_2CH_3), 3.03 (br s, 1 H, OH), 4.43 (dt, $J^{1,2}$ = 15.9 Hz, $J^{1,3}$ = 1.4 Hz, 1 H, C*H*=CHCH_2), 4.88 (q, *J* = 7.2 Hz, 1 H, C*H*CH_3), 4.89 (s, 1 H, C*H*Ph), 4.97 (s, 1 H, C*H*OH), 6.10 (dt, *J* = 15.9, 6.6 Hz, 1 H, CH=C*H*CH_2), 6.95–6.99 (m, 2 H, C_6H_5), 7.1–7.4 (m, 13 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.1, 19.8, 25.6, 53.1, 59.8, 69.2, 75.8, 123.4, 127.2, 127.4, 127.5, 127.6, 127.7, 127.8, 127.9, 128.3, 128.5, 136.2, 136.4, 139.7, 140.1, 170.6.

LC-ESI-MS (R_t 21.40 min): *m*/*z* = 394 (M – H_2O + 1), 412 (M + 1), 434 (M + Na).

Anal. Calcd for $C_{28}H_{29}NO_2$: C, 81.72; H, 7.10; N, 3.40. Found: C, 81.70; H, 7.09; N, 3.42.

(1′*S*,1″*S*,3*R*,4*S*)-4h
Yellow oil; $[\alpha]_D$ +47 (*c* = 1.0, $CHCl_3$).

IR (neat): 3423, 3065, 3032, 2972, 2932, 1719, 1646, 1494, 1461, 1381, 1208, 1043, 1023 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.82 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.36 (d, 3 H, *J* = 7.0 Hz, $CHCH_3$), 1.8–2.0 (m, 2 H, CH_2CH_3), 2.90 (br s, 1 H, OH), 4.14 (q, 1 H, *J* = 7.0 Hz, C*H*CH_3), 4.49 (s, 1 H, C*H*Ph), 4.66 (dt, $J^{1,2}$ = 15.8 Hz, $J^{1,3}$ = 1.4 Hz, 1 H, C*H*=CHCH_2), 4.90 (br s, 1 H, C*H*OH), 6.03 (dt, *J* = 15.8, 6.6 Hz, 1 H, CH=C*H*CH_2), 7.0–7.5 (m, 15 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.1, 19.2, 25.7, 52.6, 61.1, 68.7, 76.2, 121.6, 127.3, 127.4, 127.5, 127.6, 127.7, 127.8, 127.9, 128.3, 128.4, 136.1, 137.1, 139.2, 139.7, 170.3.

LC-ESI-MS (R_t 21.20 min): *m*/*z* = 394 (M – H_2O + 1), 412 (M + 1), 434 (M + Na).

Anal. Calcd for $C_{28}H_{29}NO_2$: C, 81.72; H, 7.10; N, 3.40. Found: C, 81.73; H, 7.10; N, 3.42.

3-But-1-enyl-3-(hydroxyphenylmethyl)-4-phenyl-1-(1-phenylethyl)azetidin-2-ones
(1′*S*,1″*R*,3*S*,4*R*)-3i
Yield: 72%; yellow oil; dr = 50:50; $[\alpha]_D$ –82.1 (*c* = 1.0, $CHCl_3$).

IR (neat): 3405 3063 3028 2969 2927 1731 1649 1566 1495 1454 1378 cm^{-1}.

^{1}H NMR ($CDCl_3$): δ = 0.83 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.82 (d, *J* = 7.3 Hz, 3 H, $CHCH_3$), 1.8–2.0 (m, 2 H, CH_2CH_3), 2.87 (br s, 1 H, OH), 4.45 (d, *J* = 15.7 Hz, 1 H, C*H*=CHCH_2), 4.47 (q, *J* = 7.3 Hz, 1 H, C*H*CH_3), 4.91 (s, 1 H, C*H*Ph); 5.03 (br s, 1 H, C*H*OH), 6.13 (dt, *J* = 15.7, 6.6 Hz, 1 H, CH=C*H*CH_2), 6.8–6.9 (m, 2 H, C_6H_5), 7.1–7.5 (m, 13 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.2, 20.0, 25.7, 54.0, 59.2, 69.5, 75.8, 123.3, 127.0, 127.3, 127.5, 127.8, 127.9, 128.0, 128.3, 128.4, 128.5, 135.3, 136.3, 139.8, 141.3, 170.3.

LC-ESI-MS (R_t 15.0 min): *m/z* = 394 (M – H_2O + 1), 412 (M + 1), 434 (M + Na).

Anal. Calcd for $C_{28}H_{29}NO_2$: C, 81.72; H, 7.10; N, 3.40. Found: C, 81.72; H, 7.12; N, 3.39.

(1′*S*,1″*S*,3*S*,4*R*)-4i
Yellow oil; $[\alpha]_D$ –18.2 (*c* = 0.5 $CHCl_3$).

HPLC-MS (R_t 14.8 min: *m/z* = 412 (M + 1), 434 (M + Na).

IR (neat): 3422 3063 3028 2957 2933 1731 1708 1649 1542 1448 cm^{-1}.

^1H NMR ($CDCl_3$): δ = 0.81 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.76 (d, *J* = 6.9 Hz, 3 H, $CHCH_3$), 1.85–1.95 (m, 2 H, CH_2CH_3), 2.76 (br s, 1 H, OH), 4.35 (q, *J* = 6.9 Hz, 1 H, C*H*CH_3), 4.56 (s, 1 H, C*H*Ph), 4.67 (d, *J* = 15.8 Hz, 1 H, C*H*=$CHCH_2$), 5.07 (br s, 1 H, C*H*OH), 6.04 (dt, *J* = 15.8, 6.6 Hz, 1 H, CH=C*H*CH_2), 6.90–6.93 (m, 2 H, C_6H_5), 6.91–7.07 (m, 2 H, C_6H_5) 7.2-7.5 (m, 11 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 13.2, 19.9, 25.8, 54.3, 60.8, 68.8, 76.6, 121.5, 126.8, 127.4, 127.5, 127.6, 127.8, 128.0, 128.1, 128.2, 128.3, 128.6, 128.7, 135.0, 137.2, 139.3, 141.1, 170.1.

LC-ESI-MS (R_t 14.8 min): *m/z* = 394 (M – H_2O + 1), 412 (M + 1), 434 (M + Na).

Anal. Calcd for $C_{28}H_{29}NO_2$: C, 81.72; H, 7.10; N, 3.40. Found: C, 81.71; H, 7.07; N, 3.41.

Reaction of 1 with Nitrile Derivatives or Acyl Chlorides; General Procedure
To a stirred solution of **1** (1 mmol) in anhyd THF (5 mL), were added zinc powder (1.5 mmol, 1.5 equiv, 65 mg), the nitrile derivative or acyl chloride (2 mmol, 2 equiv) under N_2 at r.t. The reaction was monitored by TLC and stopped after 2 h by adding H_2O (2 mL). The mixture was then filtered on a Celite pad to remove zinc powder and the filtrate was concentrated under reduced pressure. The residue, diluted with CH_2Cl_2 (10 mL) was washed with H_2O (2 ×). The organic layer was separated, dried (Na_2SO_4) and the solvent was removed on a rotary evaporator. Compound **5** or **6**, was isolated by flash chromatography on silica gel (cyclohexane–EtOAc, 9:1).

3-Benzoyl-1-benzyl-3-but-1-enyl-4-phenylazetidin-2-one (5b)
Yield: >95%; yellow oil.

IR (neat): 2957, 2916, 2845, 1749, 1672, 1649, 1466, 1260, 1089 1024 cm^{-1}.

^1H NMR ($CDCl_3$): δ = 0.71 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.7–1.9 (m, 2 H, CH_2CH_3), 4.02 (d, *J* = 15.0 Hz, 1 H, PhCH_2), 4.98 (d, *J* = 15.0 Hz, 1 H, PhCH_2), 5.07 (dt, $J^{1,2}$ = 15.9 Hz, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=$CHCH_2$), 5.41 (s, 1 H, C*H*Ph), 5.95 (dt, *J* = 15.9, 6.6 Hz, 1 H, C*H*=$CHCH_2$), 7.1–7.3 (m, 2 H, C_6H_5), 7.3–7.5 (m, 11 H, C_6H_5), 8.2–8.3 (m, 2 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 12.8, 25.7, 44.5, 61.7, 78.2, 122.5, 126.1, 127.7, 127.8, 128.1, 128.2, 128.3, 128.4, 128.5, 128.6, 128.7, 128.9, 130.7, 133.2, 134.3, 134.8, 136.6, 138.8, 165.6, 194.3.

LC-ESI-MS (R_t 26.17): *m/z* = 396 (M + 1), 418 (M + Na), 434 (M + K).

Anal. Calcd for $C_{27}H_{25}NO_2$: C, 82.00; H, 6.37; N, 3.54. Found: C, 82.01; H, 6.36; N, 3.54.

3-Benzoyl-3-but-1-enyl-4-phenyl-1-(1′-phenylethyl)azetidin-2-one (5i)
Yield: 60%; colorless oil; $[\alpha]_D$ +117.8 (*c* = 0.5 $CHCl_3$).

IR (neat): 3058, 3032, 2952, 2932, 1759, 1666, 1593, 1454, 1242, 1182 cm^{-1}.

^1H NMR ($CDCl_3$): δ = 0.68 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.89 (d, *J* = 7.2 Hz, 3 H, $CHCH_3$), 1.7–1.9 (m, 2 H, CH_2CH_3), 4.48 (q, *J* = 7.2 Hz, 1 H, C*H*CH_3), 5.04 (dt, $J^{1,2}$ = 15.8 Hz, $J^{1,3}$ = 1.4 Hz, 1 H, C*H*=$CHCH_2$), 5.33 (s, 1 H, C*H*Ph), 5.88 (dt, *J* = 15.8, 6.6 Hz, 1 H, CH=C*H*CH_2), 7.10–7.40 (m, 10 H, C_6H_5), 7.42–7.60 (m, 3 H, C_6H_5), 8.25 (d, *J* = 7.0 Hz, 2 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ = 12.7, 20.0, 25.6, 54.6, 61.5, 77.4, 122.6, 126.7, 127.6, 128.0, 128.1, 128.4, 128.6,130.7, 133.1, 134.7, 134.9, 138.6, 140.9, 165.6, 194.4.

LC-ESI-MS (R_t 28.00 min): *m/z* = 410 (M + 1), 432 (M + Na), 841 (2 M + Na).

Anal. Calcd for $C_{28}H_{27}NO_2$: C, 82.12; H, 6.65; N, 3.42. Found: C, 82.14; H, 6.66; N, 3.43.

3-But-1-enyl-3-phenylacetyl-4-phenyl-1-(1′-phenylethyl)azetidin-2-one (6b)
Yield: 60%; colorless oil.

IR (neat): 3063 3034 2957 2922 1761 1713 1501 1460 1401 1100 cm^{-1}.

^1H NMR ($CDCl_3$): δ= 0.83 (t, *J* = 7.5 Hz, 3 H, CH_2CH_3), 1.80–2.00 (m, 2 H, CH_2CH_3), 3.96 (d, *J* = 14.9 Hz, 1 H, PhCH_2N), 4.04 (s, 2 H, PhCH_2CO), 4.84 (dt, $J^{1,2}$ = 15.8, $J^{1,3}$ = 1.5 Hz, 1 H, C*H*=$CHCH_2$), 4.93 (d, *J* = 14.9 Hz, 1 H, PhCH_2N), 5.08 (s,1 H, C*H*Ph), 6.20 (dt, *J* = 15.8, 6.6 Hz, 1 H, CH=C*H*CH_2), 7.00–7.20 (m, 4 H, C_6H_5), 7.22–7.30 (m, 2 H, C_6H_5), 7.31–7.50 (m, 9 H, C_6H_5).

^{13}C NMR ($CDCl_3$): δ= 12.9, 25.7, 44.4, 46.3, 61.0, 78.3, 120.4, 126.9, 127.8, 127.9, 128.0, 128.1, 128.2, 128.3, 128.4, 128.8, 129.1, 129.7, 133.6, 134.0, 134.8, 139.2, 165.8, 202.0.

LC-ESI-MS (R_t 23.50 min): *m/z* = 410 (M + 1), 432 (M + Na).

Anal. Calcd for $C_{28}H_{27}NO_2$: C, 82.12; H, 6.65; N, 3.42. Found: C, 82.10; H, 6.64; N, 3.41.

Acknowledgment

We thank MURST (Cofin 2002 and FIRB 2001), ISOF-CNR and the University of Bologna (funds for selected topics) for financial support.

References

(1) (a) Knochel, P.; Millot, N.; Rodriguez, A. L.; Tucker, C. E. *Org. React.* **2001**, *58*, 417. (b) Knochel, P.; Singer, R. D. *Chem. Rev.* **1993**, *93*, 2117.
(2) (a) Masuyama, Y. *Rec. Res. Dev. Org. Chem.* **2000**, *4*, 373. (b) Luche, J.-L.; Sarandeses, L. A. In *Organozinc Reagents – A Practical Approach*; Knochel, P.; Jones, P., Eds.; Oxford University Press: Oxford, **1999**, 307–323. (c) Alonso, F.; Yus, M. *Rec. Res. Dev. Org. Chem.* **1997**, *1*, 397. (d) Tagliavini, E.; Trombini, C.; Umani-Ronchi, A. In *Advances in Carbanion Chemistry*, Vol. 2; Snieckus, V., Ed.; Jai Press Inc: Greenwich, **1996**, 111–146. (e) Rathke, M. K.; Weipert, P. In *Comprehensive Organic Synthesis*, Vol. 2; Trost, B. M., Ed.; Pergamon: Oxford, **1991**, 277–299. (f) Fürstner, A. *Synthesis* **1989**, 571. (g) Heathcock, C. H. In *Asymmetric Synthesis*, Vol. 3; Morrison, J. D., Ed.; Academic Press: New York, **1984**, 144–152.

(3) (a) Clader, J. *Science* **2004**, *303*, 1201. (b) Singh, G. S. *Mini Rev. Med. Chem.* **2004**, *4*, 69. (c) Wu, G. G. *Org. Process Res. Dev.* **2000**, *4*, 298. (d) McKittrick, B. A.; Ma, K.; Huie, K.; Yumibe, N.; Davis, H. Jr.; Clader, J. W.; Czarniecki, M.; McPhail, A. T. *J. Med. Chem.* **1998**, *41*, 752. (e) Burnett, D. A.; Caplen, M. A.; Davis, H. Jr.; Burrier, R. E.; Clader, J. W. *J. Med. Chem.* **1994**, *37*, 1733.

(4) (a) Alcaide, B.; Almendros, P.; Aragoncillo, C.; Rodriguez-Acebes, R. *J. Org. Chem.* **2004**, *69*, 826. (b) Alcaide, B.; Almendros, P.; Aragoncillo, C.; Rodriguez-Acebes, R. *Synthesis* **2003**, 1163. (c) Alcaide, B.; Almendros, P.; Aragoncillo, C.; Rodriguez-Acebes, R. *J. Org. Chem.* **2001**, *66*, 5208. (d) Paquette, L. A.; Rothhaar, R. R.; Isaac, M.; Rogers, L. M.; Rogers, R. D. *J. Org. Chem.* **1998**, *63*, 5463. (e) Jayaraman, M.; Batista, M. T.; Manhas, M. S.; Bose, A. K. *Heterocycles* **1998**, *49*, 97. (f) Jayaraman, M.; Manhas, M. S.; Bose, A. K. *Tetrahedron Lett.* **1997**, *38*, 709. (g) Palomo, C.; Aizpurua, J. M.; Lopez, M. C.; Aurrekoetxea, N.; Oiarbide, M. *Tetrahedron Lett.* **1990**, *31*, 6425.

(5) Palomo, C.; Aizpurua, J. M.; Ganboa, I.; Oiarbide, M. *Eur. J. Org. Chem.* **1999**, 3223.

(6) (a) Cardillo, G.; Fabbroni, S.; Gentilucci, L.; Perciaccante, R.; Piccinelli, F.; Tolomelli, A. *Tetrahedron* **2004**, *60*, 5031. (b) Cardillo, G.; Fabbroni, S.; Gentilucci, L.; Perciaccante, R.; Tolomelli, A. *Tetrahedron: Asymmetry* **2004**, *15*, 593. (c) Cardillo, G.; De Simone, A.; Mingardi, A.; Tomasini, C. *Synlett* **1995**, 1131.

(7) (a) Manhas, M. S.; Banik, B. K.; Mathur, A.; Vincent, J. E.; Bose, A. K. *Tetrahedron* **2000**, *56*, 5587. (b) Manhas, M. S.; Ghosh, M.; Bose, A. K. *J. Org. Chem.* **1990**, *55*, 575. (c) Zamboni, R.; Just, G. *Can. J. Chem.* **1979**, *57*, 1945. (d) Bose, A. K.; Spiegelmann, G.; Manhas, M. S. *Tetrahedron Lett.* **1971**, 3167.

(8) (a) Bari, S. S.; Venugopalan, P.; Arora, R. *Tetrahedron Lett.* **2003**, *44*, 895. (b) Kingston, D. G. I. *Chem. Commun.* **2001**, 867. (c) Bandini, E.; Favi, G.; Martelli, G.; Panunzio, M.; Piersanti, G. *Org. Lett.* **2000**, *2*, 1077. (d) Kende, A. S.; Liu, K.; Kaldor, I.; Dorey, G.; Koch, K. *J. Am. Chem. Soc.* **1995**, *117*, 8258. (e) Nicolaou, K. C.; Doi, W. M.; Guy, R. K. *Angew. Chem., Int. Ed. Engl.* **1994**, *33*, 15. (f) Ojima, I.; Habus, I.; Zhao, M.; Zucco, M.; Park, Y. H.; Sun, C. M.; Brigaud, T. *Tetrahedron* **1992**, *48*, 6985. (g) Duran, F.; G hosez, L. *Tetrahedron Lett.* **1970**, *11*, 245.

(9) The reactions were performed on racemic mixtures. For the sake of simplicity only one enantiomer has been reported in the schemes.

(10) (a) Freihammer, P. M.; Detty, M. R. *J. Org. Chem.* **2000**, *65*, 7203. (b) Durst, T.; Sharma, M. K.; Gabe, E. J.; Lee, F. L. *J. Org. Chem.* **1990**, *55*, 5525.

(11) Detailed X-ray data for **3a** and **3b** are reported in the experimental section. CCDC number for **3a** is 245012, CCDC number for **3b** is 245013.

(12) (a) Lindstroem, U. M. *Chem. Rev.* **2002**, *102*, 2751. (b) Kobayashi, S.; Manabe, K. *Acc. Chem. Res.* **2002**, *35*, 209. (c) Fringuelli, F.; Piermatti, O.; Pizzo, F.; Vaccaro, L. *Eur. J. Org. Chem.* **2001**, *3*, 439. (d) Dias, L. C. *Curr. Org. Chem.* **2000**, *4*, 305. (e) Laszlo, P.; Teston, M. *J. Am. Chem. Soc.* **1990**, *112*, 8750.

(13) (a) Kobayashi, S.; Busujima, T.; Nagayama, S. *Chem. Eur. J.* **2000**, *6*, 3491. (b) Ferraris, D.; Drury, W. J. III; Cox, C.; Lectka, T. *J. Org. Chem.* **1998**, *63*, 4568. (c) Pearson, R. G. *J. Am. Chem. Soc.* **1963**, *85*, 3533.

(14) (a) Dilman, A. D.; Ioffe, S. L. *Chem. Rev.* **2003**, *103*, 733. (b) Bongini, A.; Cardillo, G.; Mingardi, A.; Tomasini, C. *Tetrahedron: Asymmetry* **1996**, *7*, 1457. (c) Lipshutz, B. H.; Dimock, S. H.; James, B. *J. Am. Chem. Soc.* **1993**, *115*, 9283. (d) Lipshutz, B. H.; James, B. *Tetrahedron Lett.* **1993**, *34*, 6689.

(15) Cainelli, G.; Cardillo, G.; Orena, M.; Sandri, S. *J. Am. Chem. Soc.* **1976**, *98*, 6737.

(16) (a) Iwai, T.; Ito, T.; Mizuno, T.; Ishino, Y. *Tetrahedron Lett.* **2004**, *45*, 1083. (b) Lee, A. S.-Y.; Lin, L.-S. *Tetrahedron Lett.* **2000**, *41*, 8803.